AUDIO UNIT [WITHOUT CENTER DISPLAY]

id0920z9010200

Note
- iPod is a registered trademark of Apple Inc.

Purpose
- The audio unit controls the entertainment system-related parts.

Function
Display function
- Displays audio information such as radio and CD related information in the audio unit display.

Memory function
- The audio unit stores the following information.
  - — DTC
  - — Radio frequencies stored in preset
  - — Sound quality settings
    - AF (Setting Alternative Frequency (AF))
    - REG (Setting Regional Programme (REG))
    - ALC (auto volume adjustment)
    - BASS (low range adjustment)
    - TREB (high range adjustment)
    - FADE (front/back volume adjustment)
    - BAL (left/right volume adjustment
    - BEEP (operation sound setting)
    - BT SETUP
  - — Clock setting
    - 12Hr/24Hr (time display setting)
  - — Volume setting

Note
- The sound quality, clock, and volume settings stored in the audio unit are erased if the battery cables are disconnected.

ALC (auto level control) function
- Compensates for the passengers' sense of a lack of audio volume which occurs as a result of external noise while driving.

Sound quality/time adjustment function
- The sound quality of the audio unit and the time displayed on the audio unit display can be adjusted by rotating the POWER/VOLUME button after selecting the MENU button of the audio unit.

Mode	Direction in which POWER/VOLUME button is rotated	
	Left	Right
AF (Setting Alternative Frequency (AF))*1	Off	On
REG (Setting Regional Programme (REG))*1	Off	On
ALC (auto volume adjustment)	Volume change low	Volume change high
BASS (low range adjustment)	Low bass range	High bass range
TREB (high range adjustment)	Low high range	High high range

Mode	Direction in which POWER/VOLUME button is rotated	
	Left	Right
FADE (front/ back volume adjustment)	Rear speaker low volume	Front speaker low volume
BAL (left/right volume adjustment	Right speaker low volume	Left speaker low volume
BEEP (operation sound setting)	BEEP sound off	BEEP sound on
BT SETUP	Select mode	
12Hr/24Hr (time display setting)	12 Hr display (flashes)	24 Hr display (flashes)

*1 : European (L.H.D. U.K.) specs.

TA (traffic announcement) function

- When traffic information is received while listening in the following modes, the audio switches automatically to the traffic information.
  - FM radio
  - CD
  - AUX
  - USB (iPod)
  - Bluetooth® audio
- When the traffic information has finished, the audio unit returns to the mode that was being listened to before the traffic information broadcast.
- If the TA function is set to ON, "TA" is displayed in the audio unit display regardless of traffic information reception.

Note

- If traffic information is received while the TA function is set to OFF, "TA" is displayed in the audio unit display. If the TA button on the audio panel is selected during "TA" display, announcement of traffic information starts.
- If the following operations are performed during a traffic information announcement, the traffic information announcement is cancelled.
  - Turn the audio unit power off
  - Switch to AM radio by operating FM/AM/CLOCK button
  - Turn TA function off by operating TA button

On-board diagnostic function

- The audio unit has an on-board diagnostic function to facilitate system diagnosis. DTCs are stored in the event of a malfunction occurring. For details, refer to the DIAGNOSTIC SYSTEM. (See ON-BOARD DIAGNOSTIC [ENTERTAINMENT SYSTEM (WITHOUT CENTER DISPLAY)].)

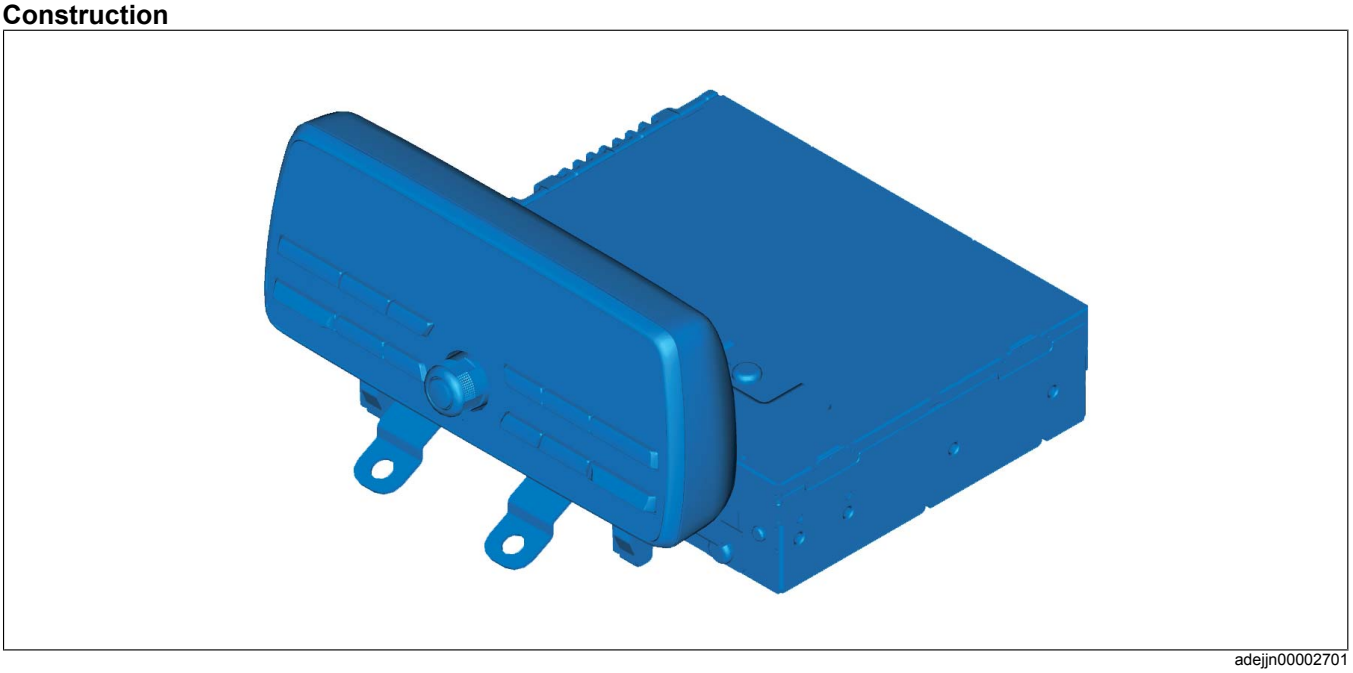
Diagnostic assist function

- A diagnostic assist function is equipped which can verify the following. For details, refer to the DIAGNOSTIC SYSTEM. (See ON-BOARD DIAGNOSTIC [ENTERTAINMENT SYSTEM (WITHOUT CENTER DISPLAY)].)
  - Connection status of related parts
  - Operation status of related parts
  - Related part specifications

Specification

Item			Specification
Rated voltage		(V)	12
Frequency range	AM	(kHz)	531—1602 (European (L.H.D. U.K.) specs.) 530—1620 (With AM frequency pitch: 5KHz pitch) 522—1629 (With AM frequency pitch: 9KHz pitch)
	FM	(MHz)	87.5—108
	LW	(kHz)	153—279
Amplifier maximum output		(W)	25×4
Output impedance		(ohm)	4

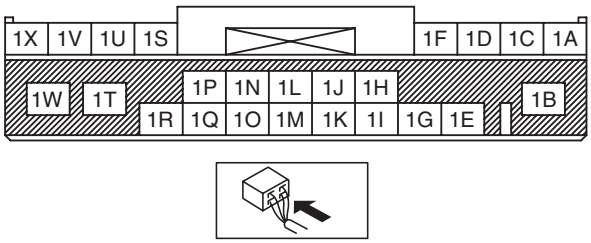
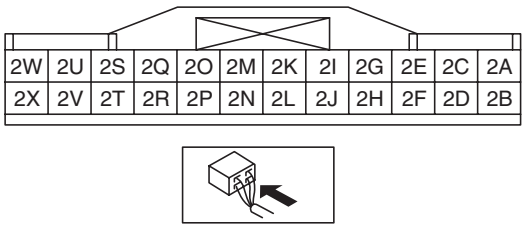
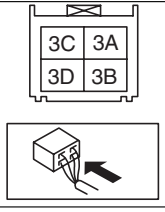
Construction



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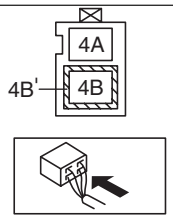
Input/output signal table

Audio unit

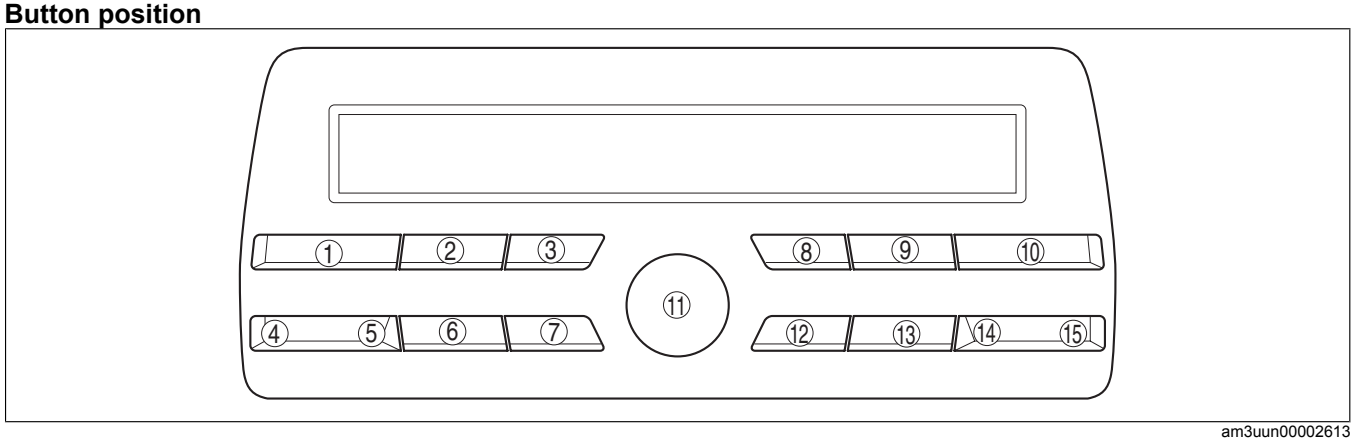
TERMINAL		Signal
	1A	Front door speaker LH (+)
	1B	B+
	1C	Front door speaker LH (-)
	1D	Front door speaker RH (+)
	1E	TNS signal input
	1F	Front door speaker RH (-)
	1G	Illumination power supply (-)
	1H	—
	1I	Vehicle speed signal input
	1J	—
	1K	—
	1L	Dimmer cancel
	1M	—
	1N	Steering switch1
	1O	MS-CAN (+)
	1P	Steering switch2
	1Q	MS-CAN (-)
	1R	ACC
	1S	Rear door speaker LH (+)
	1T	—
	1U	Rear door speaker LH (-)
	1V	Rear door speaker RH (+)
	1W	Ground
	1X	Rear door speaker RH (-)
	2A	TEL (+)*2
	2B	TEL (-)*2
	2C	Bluetooth® audio input LH (+)*2
	2D	Bluetooth® audio input LH/RH (-)*2
	2E	Bluetooth® audio input RH (+)*2
	2F	—
	2G	—
	2H	—
	2I	AUX audio input LH (+)
	2J	AUX audio input LH/RH (-)
	2K	AUX audio input RH (+)
	2L	AUX connection detection
	2M	CD audio input LH (+)*3
	2N	CD audio input LH (-)*3
	2O	CD audio input RH (+)*3
	2P	CD audio input RH (-)*3
	2Q	—
	2R	—
	2S	—
	2T	—
	2U	—
	2V	—
	2W	—
	2X	—
	3A	USB power supply ground
	3B	USB data (+)
	3C	USB power supply
	3D	USB data (-)

*2 : With Bluetooth® unit
*3 : With CD player

Antenna feeder connector

TERMINAL		Signal
	4A	AM/FM antenna amplifier power supply
	4B	Radio input
	4B'	Ground

Button position



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European (L.H.D. U.K.) specs.

No.	Button name	Function
1	Preset button 1/:00 button	Switches to radio station programmed in Preset 1.
2	Preset button 2	Switches to radio station programmed in Preset 2.
3	Preset button 3	Switches to radio station programmed in Preset 3.
4	TUNE button (down)	Changes CD tracks or switches radio stations.
5	TUNE button (up)	Changes CD tracks or switches radio stations.
6	FM/AM/CLOCK button	• Switches to radio mode. • Switches to clock adjustment mode. (When pressed continuously for 1.5 s or more)
7	MEDIA/SCAN button	• Switches the mode. • Search and selection of receivable radio station is possible.
8	Preset button 4	Switches to radio station programmed in Preset 4.
9	Preset button 5	Switches to radio station programmed in Preset 5.
10	Preset button 6	Switches to radio station programmed in Preset 6.
11	POWER/VOLUME button	• Audio power on/off and volume adjustment are possible. • Audio sound quality can be adjusted.
12	MENU button	• Switches to sound quality adjustment mode. The display switches in the following order. — AF→REG→ALC→BASS→TREB→FADE→BAL→BEEP→BT SETUP→12Hr< >24Hr
13	TA button	Switches TA (traffic announcement) on/off.

No.	Button name	Function
14	FLDR/PTY button (down)	• Switches to PTY selection mode. After selecting the FLDR/PTY button, the PTY code currently being received is displayed in the audio unit display. • When the button is long-pressed, PTY seek is performed based on the radio frequency currently being received. The seek direction is in the down direction. **Note** • PTY seek is performed until the radio station which has PTY code corresponding to the PTY code currently being selected is received. • Selects the folder to be played (MP3/WMA/AAC/CD playback).
15	FLDR/PTY button (up)	• Switches to PTY selection mode. After selecting the FLDR/PTY button, the PTY code currently being received is displayed in the audio unit display. • When the button is long-pressed, PTY seek is performed based on the radio frequency currently being received. The seek direction is in the up direction. **Note** • PTY seek is performed until the radio station which has PTY code corresponding to the PTY code currently being selected is received. • Selects the folder to be played (MP3/WMA/AAC/CD playback).

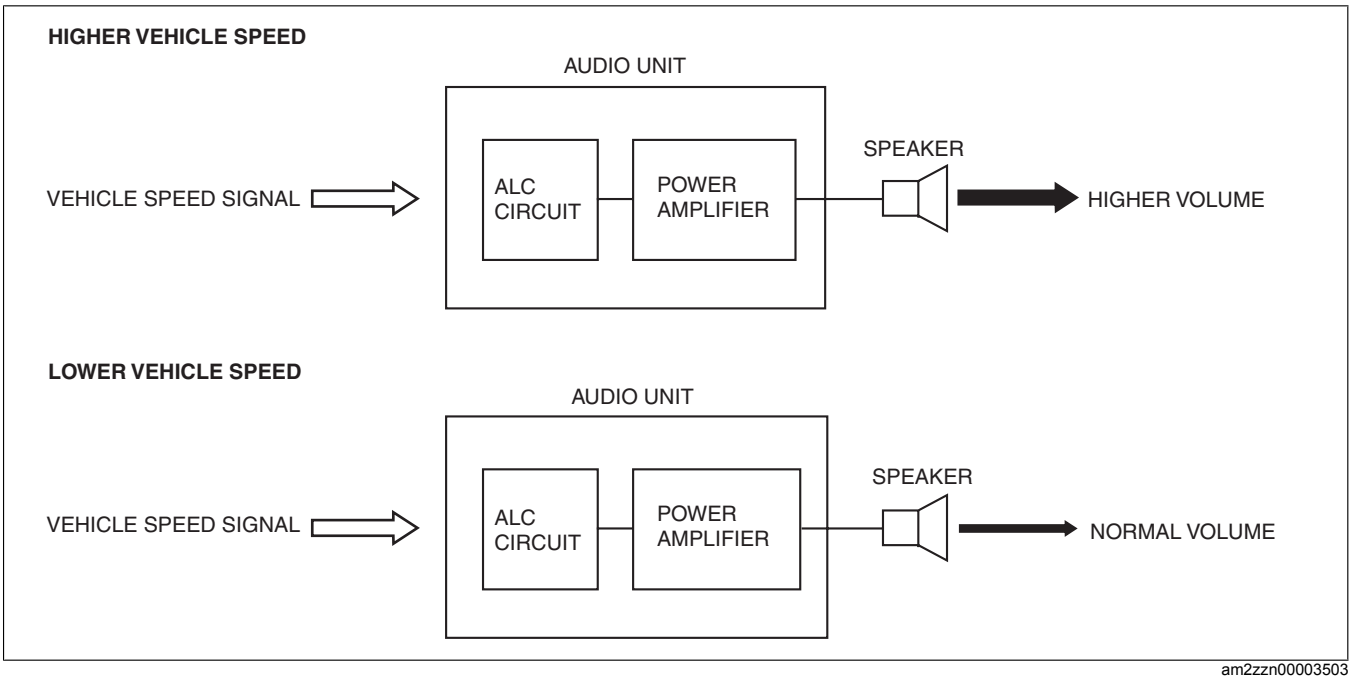
Except European (L.H.D. U.K.) specs.

No.	Button name	Function
1	Preset button 1	Switches to radio station programmed in Preset 1.
2	Preset button 2	Switches to radio station programmed in Preset 2.
3	Preset button 3	Switches to radio station programmed in Preset 3.
4	TUNE button (down)	Changes CD tracks or switches radio stations.
5	TUNE button (up)	Changes CD tracks or switches radio stations.
6	FM/AM/CLOCK button	• Switches to radio mode. • Switches to clock adjustment mode. (When pressed continuously for 1.5 s or more)
7	MEDIA/SCAN button	• Switches the mode. • Search and selection of receivable radio station is possible.
8	Preset button 4	Switches to radio station programmed in Preset 4.
9	Preset button 5	Switches to radio station programmed in Preset 5.
10	Preset button 6	Switches to radio station programmed in Preset 6.
11	POWER/VOLUME button	• Audio power on/off and volume adjustment are possible. • Audio sound quality can be adjusted.
12	MENU button	• Switches to sound quality adjustment mode. The display switches in the following order. — ALC→BASS→TREB→FADE→BAL→BEEP→BT SETUP→12Hr< >24Hr
13	AUTO-M button	Stores up to six receivable stations.
14	FOLDER button (down)	Selects folder to be played.
15	FOLDER button (up)	Selects folder to be played.

Operation

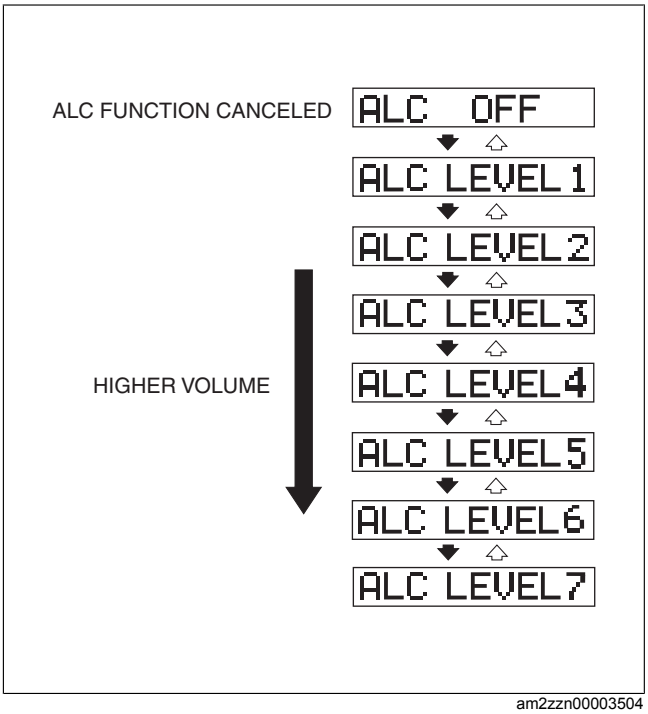
ALC (auto level control) operation

- The audio unit automatically changes the volume and audio quality based on the vehicle speed signal received from the instrument cluster.



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- The ALC function is divided into 7 levels. When the level is in the high range, the volume correction range and correction amount are large. The maximum correction amount is **14dB**. The ALC function can be stopped by selecting ALC OFF.



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Fail-safe

- Not applicable